

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2020

Mathematics

Paper 2

Ordinary Level

Time: 2 hours

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

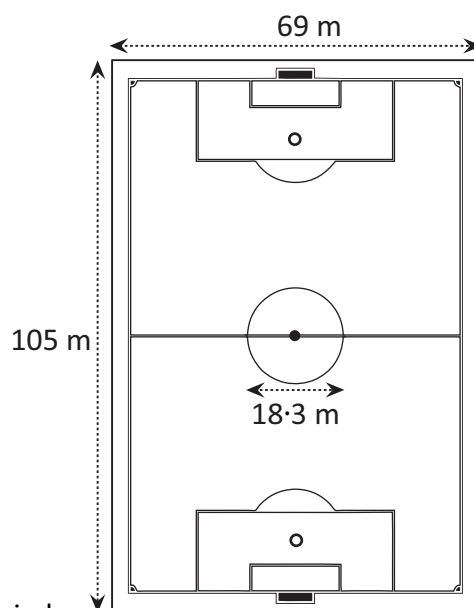
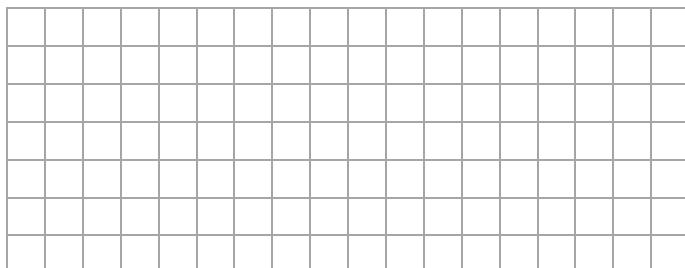
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Question 1

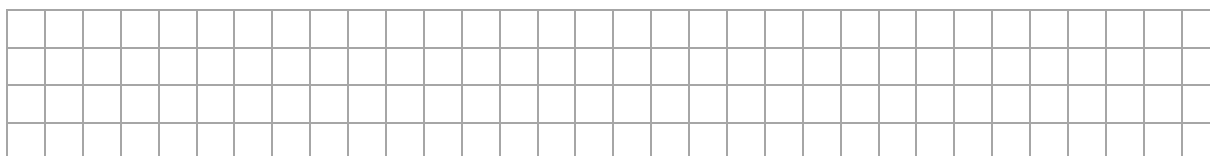
(Suggested maximum time: 10 minutes)

The Euro 2020 Football Final will take place in Wembley Stadium, London on 12th July.

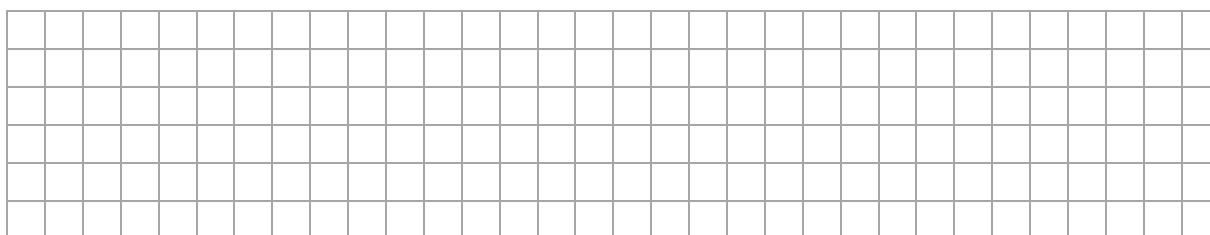
- (a) The pitch in the stadium measures 105 m long and 69 m wide. Work out the **area** of the pitch.



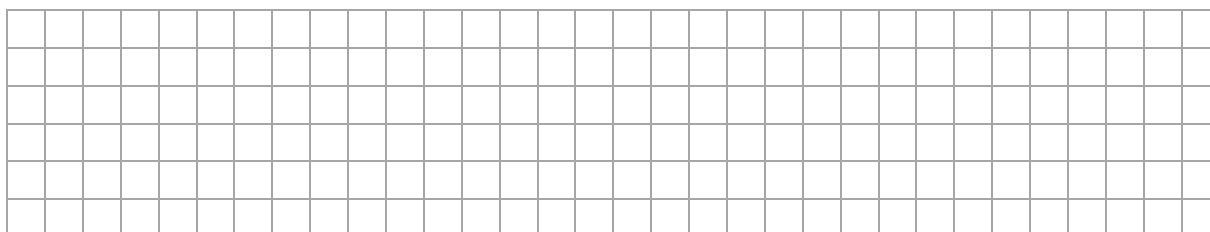
- (b) The centre circle of the pitch has a diameter of 18.3 m
(i) Write down the length of the **radius** of the centre circle.



- (ii) Work out the **area** of the centre circle.
Give your answer correct to the nearest m^2 .



- (iii) Express the area of the centre circle as a percentage of the total area of the pitch.
Give your answer correct to one decimal place.



- (c) 4 matches during the Euro 2020 football tournament final will be played in Dublin.
There will be 51 matches played in total during the tournament.
A match is picked at random.

Write down the probability that the match will be played in Dublin.

Answer:

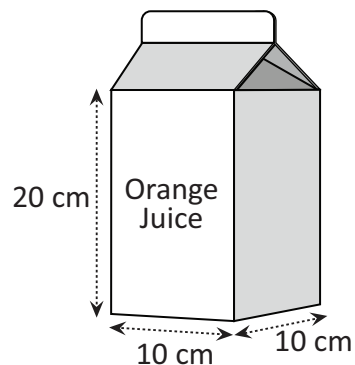
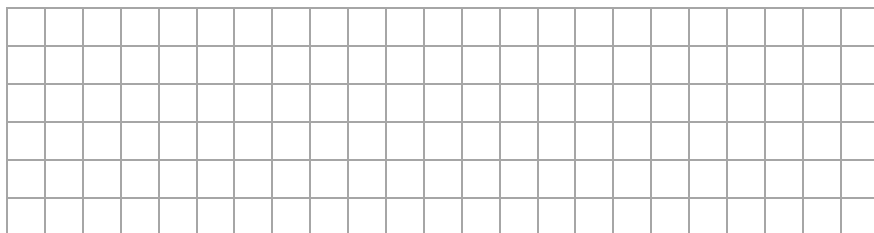
Question 2

(Suggested maximum time: 15 minutes)

- (a) A rectangular carton of orange juice has a square base with sides of length 10 cm, and a height of 20 cm.

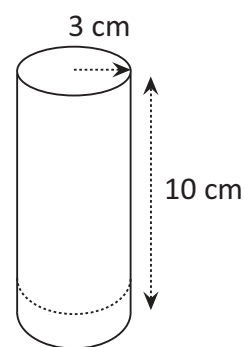
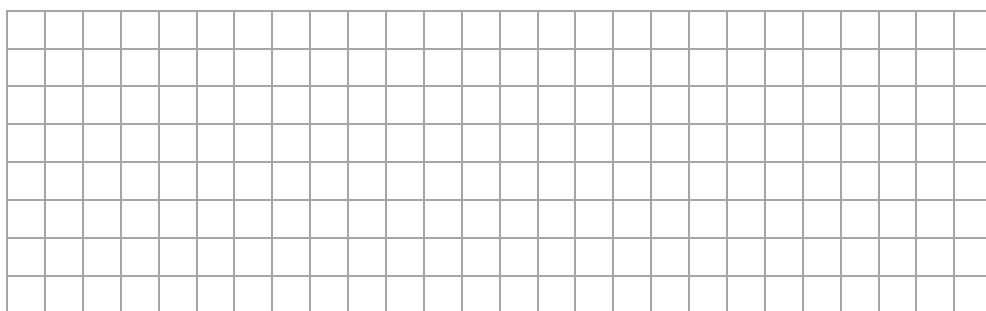
Work out the **volume** of the rectangular carton, in litres.

Note: $1000 \text{ cm}^3 = 1 \text{ litre}$.

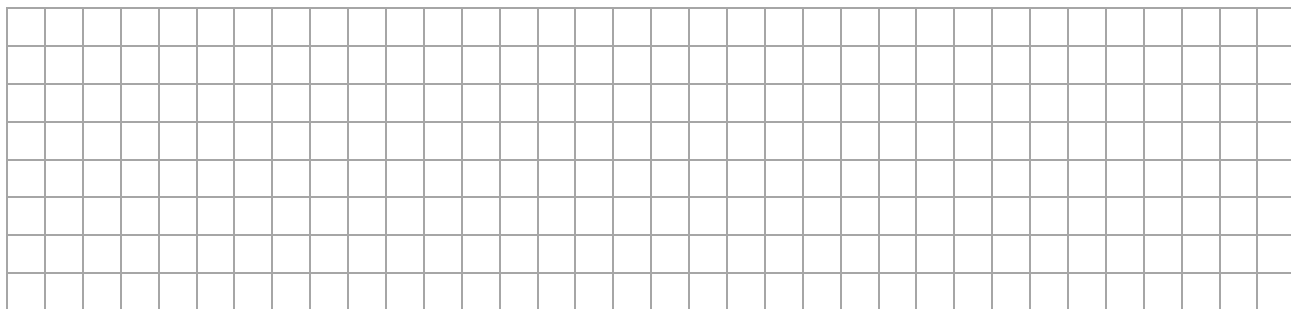


- (b) Cylindrical glasses of height 10 cm and radius 3 cm are to be filled from the carton.

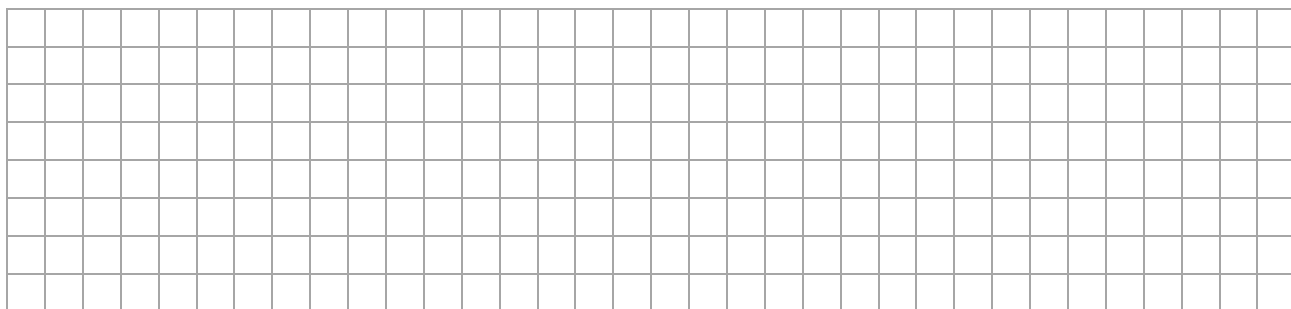
- (i) Work out the **volume** of one cylindrical glass.
Give your answer, in cm^3 , correct to one decimal place.



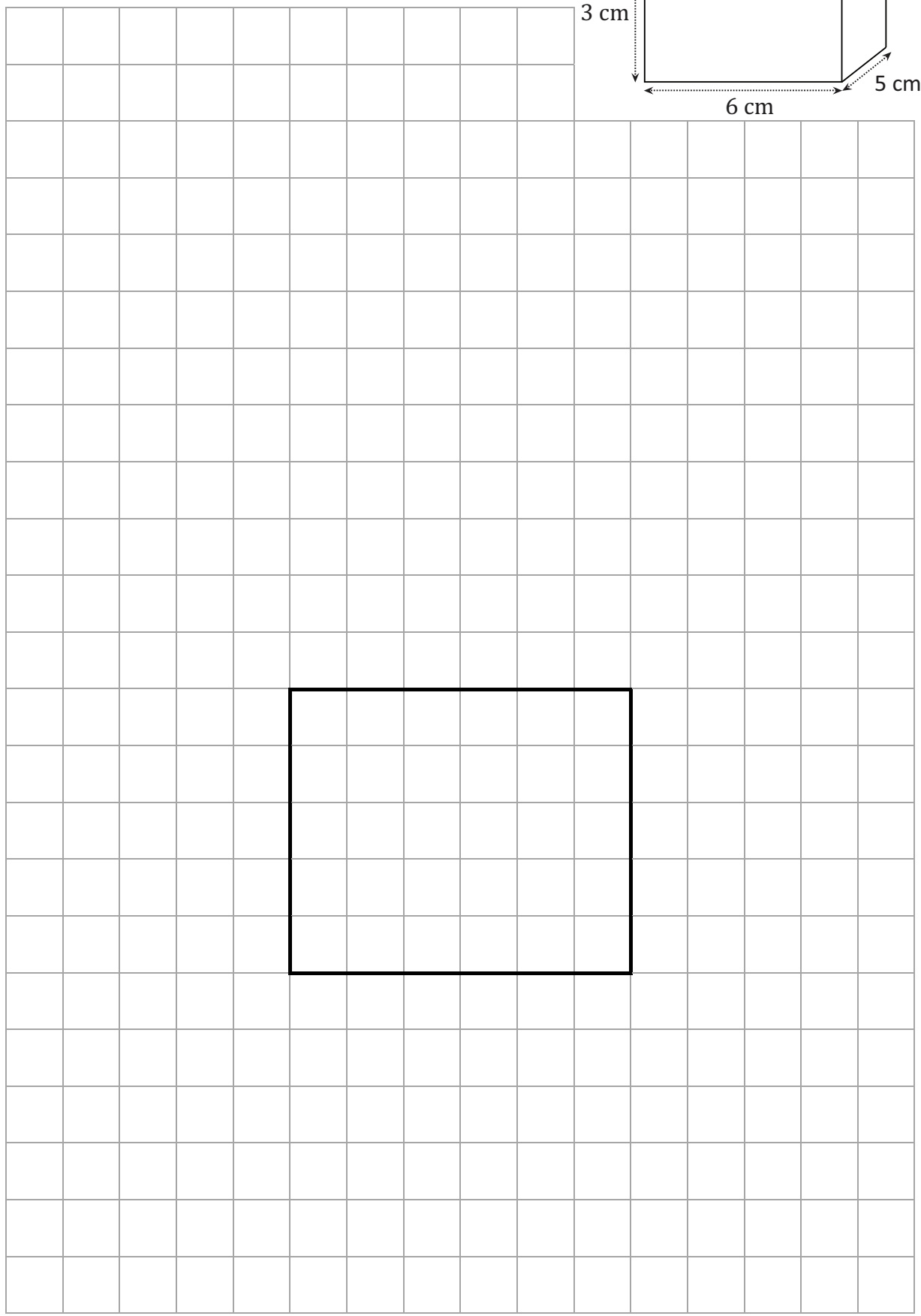
- (ii) Work out how many full glasses can be filled with orange juice from the carton.



- (iii) How much orange juice is **left over** after filling the same number of glasses as in part (ii)?
Give your answer, in cm^3 , correct to one decimal place.



- (c) A closed rectangular box has length 6 cm, width 5 cm and a height of 3 cm.
 Draw a **net** of the closed rectangular box, as accurately as you can.
 The grid below has small squares with sides of length 1 cm.
 The base of the box is done for you.



Question 3

(Suggested maximum time: 5 minutes)

A bag contains eight cards with a shape on each, as shown below.



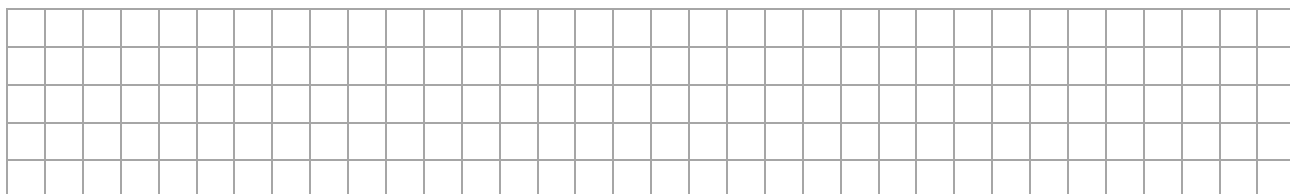
One card is picked at random from the bag.

- (a) The box below shows terms that can be used to describe the probability that an event happens.

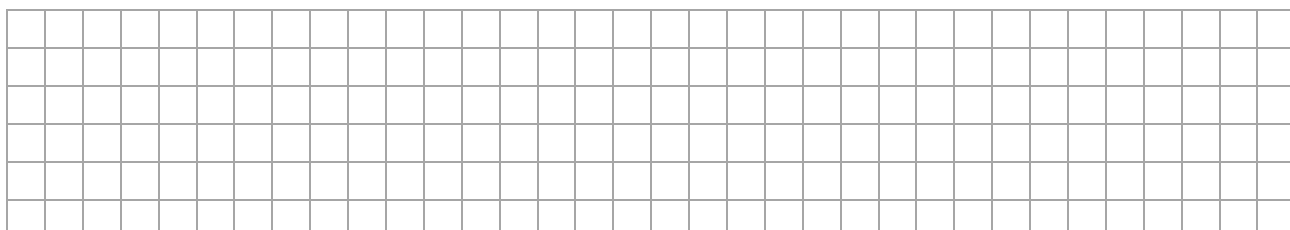
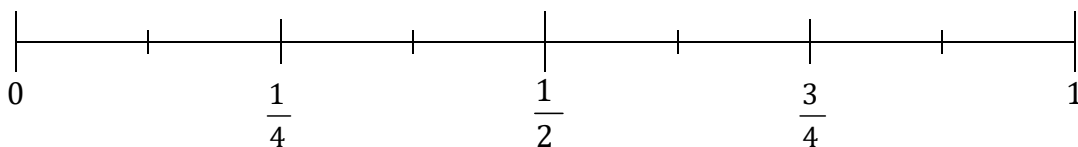
Unlikely	Certain	Likely	Impossible	Equal Chance
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For each event **A**, **B**, **C** and **D** in the table below, use one of the terms from the box to describe the probability that it happens.

Event	Description	Probability
A	The card picked has a circle on it.	
B	The card picked has a rectangle on it.	
C	The card picked has a shape on it.	
D	The card picked has a diamond on it.	



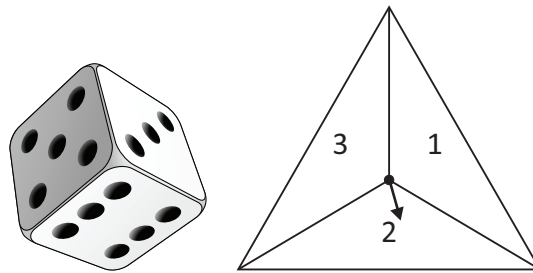
- (b) Write each of the letters **A**, **B**, **C** and **D** in the correct place on the probability scale below to show the probability of each event.



Question 4 (Suggested maximum time: 5 minutes)

Question 4 (Suggested maximum time: 5 minutes)

Fionn rolls a fair six-sided die and spins a fair spinner with three equal sectors, numbered 1 to 3 as shown below. The results of the die and spinner are then **added** together.



- (a)** Complete the table below to show all the possible outcomes.

		Die					
		1	2	3	4	5	6
Spinner	1	2					
	2						
	3		5				

- (b)** How many possible outcomes are there?

Answer:

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[illegible]

- (c) How many outcomes give an even number as a result?

Answer:

Page 10 of 10

[illegible]

- (d)** Write down the probability that the result is an even number.

Answer:

[illegible]

Question 5 (Suggested maximum time: 15 minutes)

During a survey Nick asked people to state the number of children in their families. The results are listed below.



2	3	3	0	2
1	0	5	4	2
1	3	1	1	5
4	1	2	2	0

- (a)** Write down the number of people who took part in Nick's survey. Answer:

[illegible]

- (b)** Complete the frequency table below to show the number of children in each family.

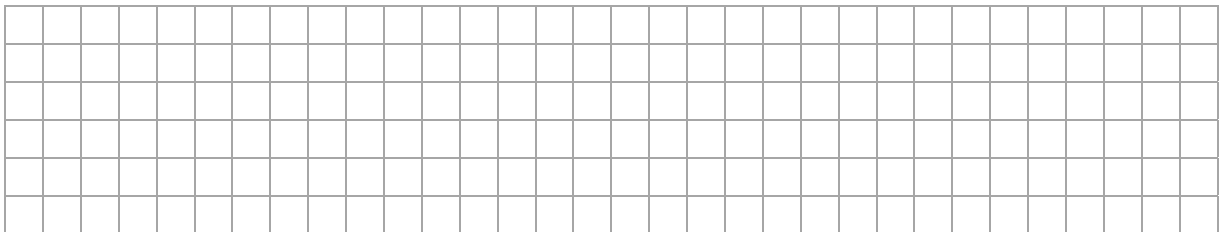
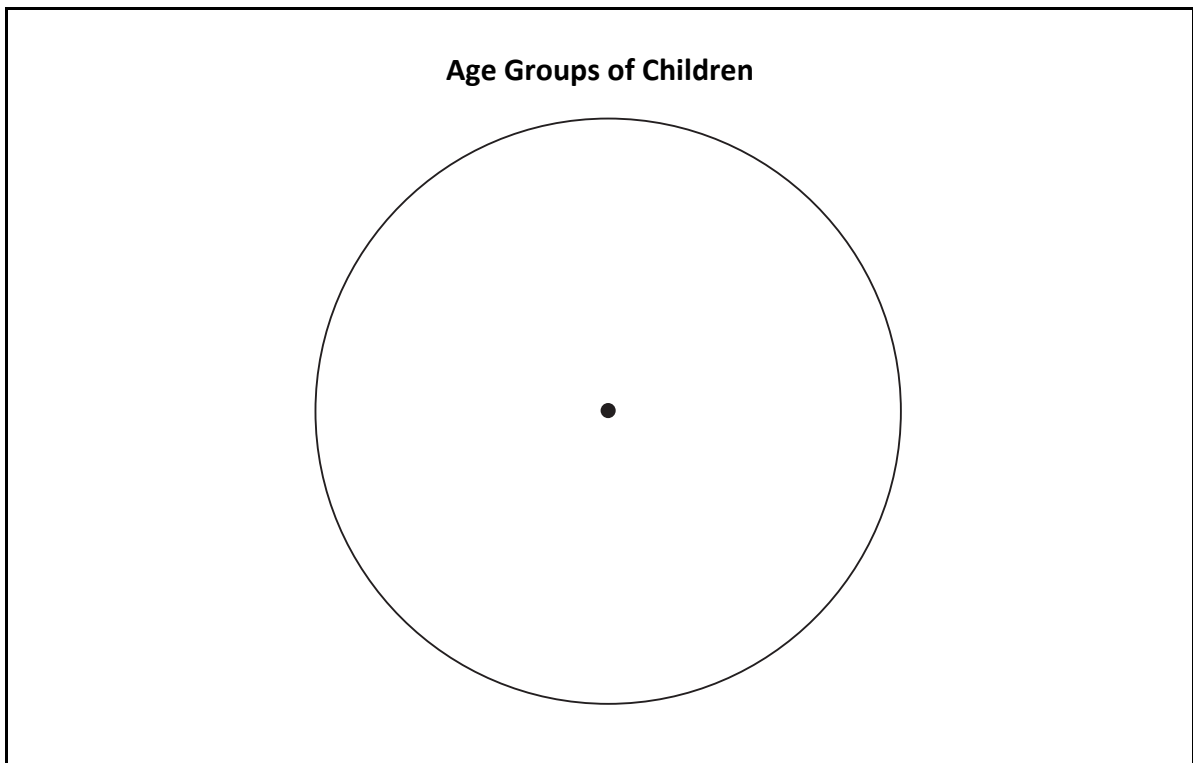
Number of children in each family	0	1	2	3	4	5
Number of families						

[illegible]

- (c)** Display this data graphically. Label your graph clearly.

[illegible]

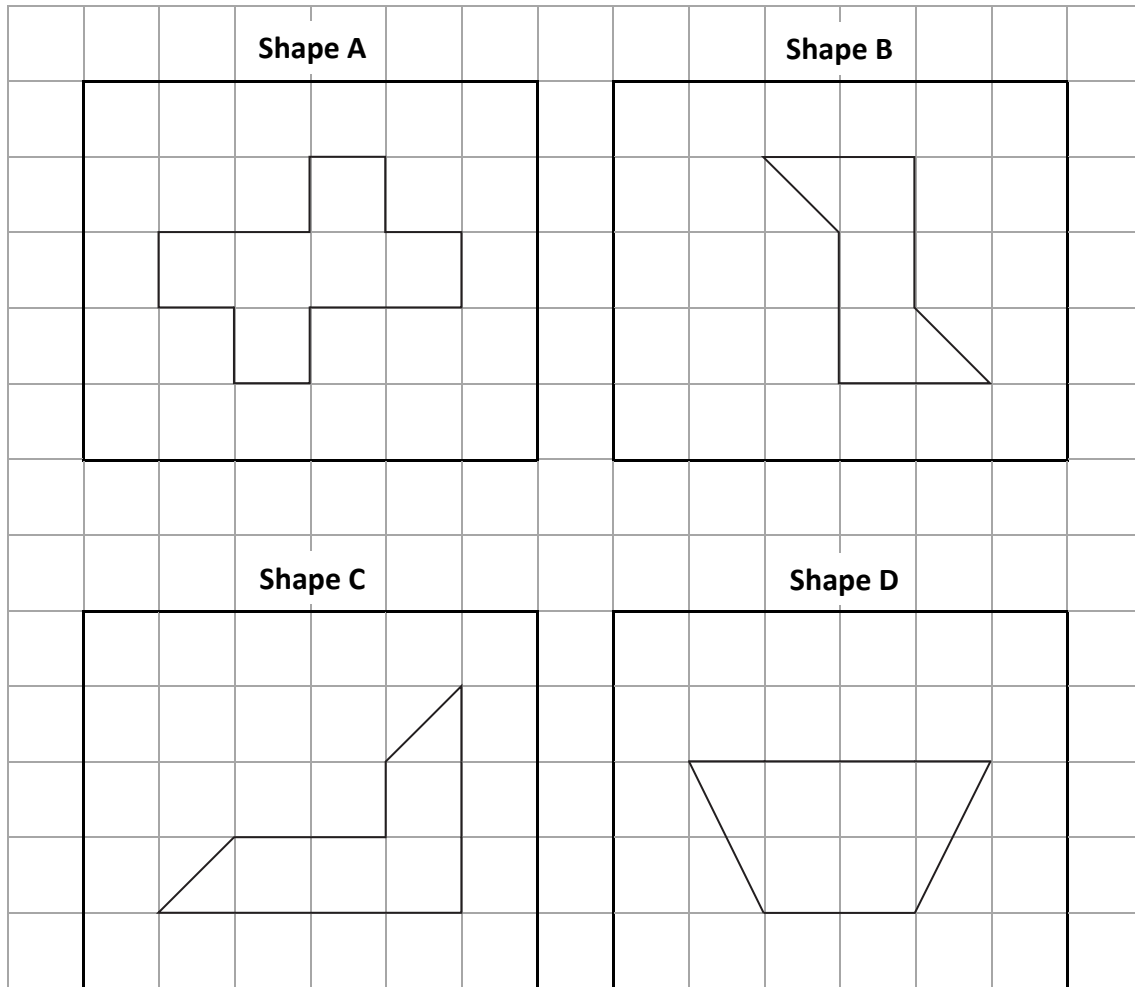
- (e) (iii) Complete the pie chart below to show the number of children in each age group.
Label each sector clearly with the age group of the children and the size of the angle.



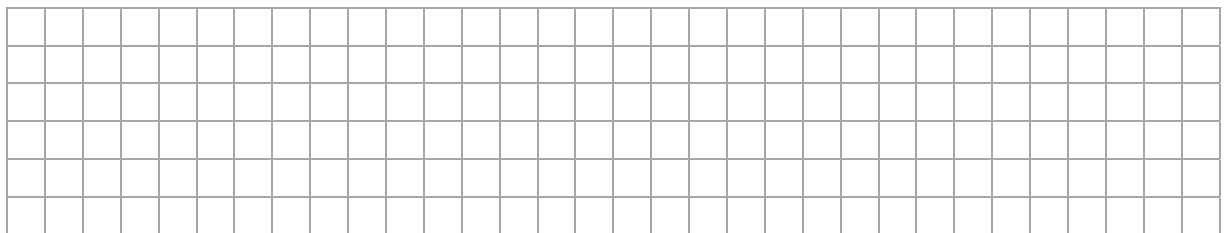
Question 6

(Suggested maximum time: 5 minutes)

Four shapes are shown below. Each small square on the grid has side of length 1 cm.



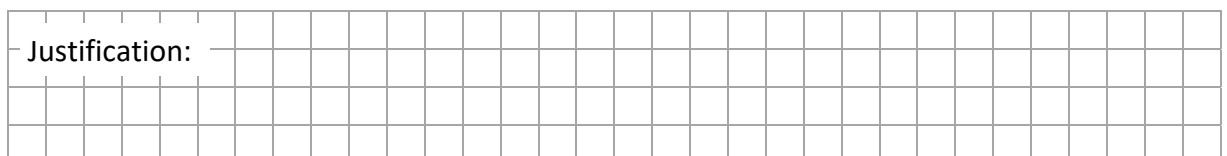
- (a) Find the length of the **perimeter** of shape A.



- (b) Which shape has the smallest **area**?

Answer:

Justify your answer.



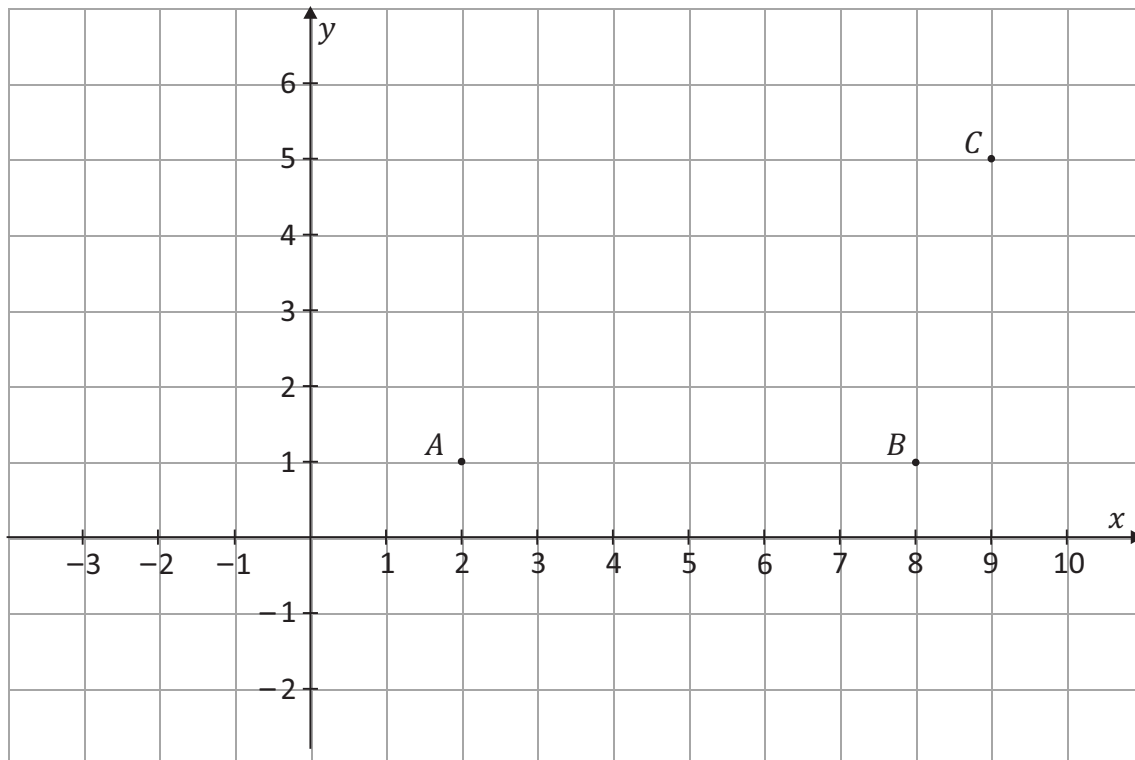
- (c) Which shape has an **axis of symmetry**?

Answer:

Question 7 (Suggested maximum time: 10 minutes)

Question 7 (Suggested maximum time: 10 minutes)

The points A , B and C are shown on the co-ordinated diagram below.



- (a)** Write down the co-ordinates of the points A , B and C .

$$A = \begin{pmatrix} & \\ & \end{pmatrix} \quad B = \begin{pmatrix} & \\ & \end{pmatrix} \quad C = \begin{pmatrix} & \\ & \end{pmatrix}$$

- (b)** D is the point $(3, 5)$.
Plot the point D on the co-ordinated diagram above.

- (c) Jamie says: "The shape formed by the points A , B , C and D is a **parallelogram**". Is he correct? Justify your answer fully. Tick (✓) **one** box only.

Yes

1

No

9

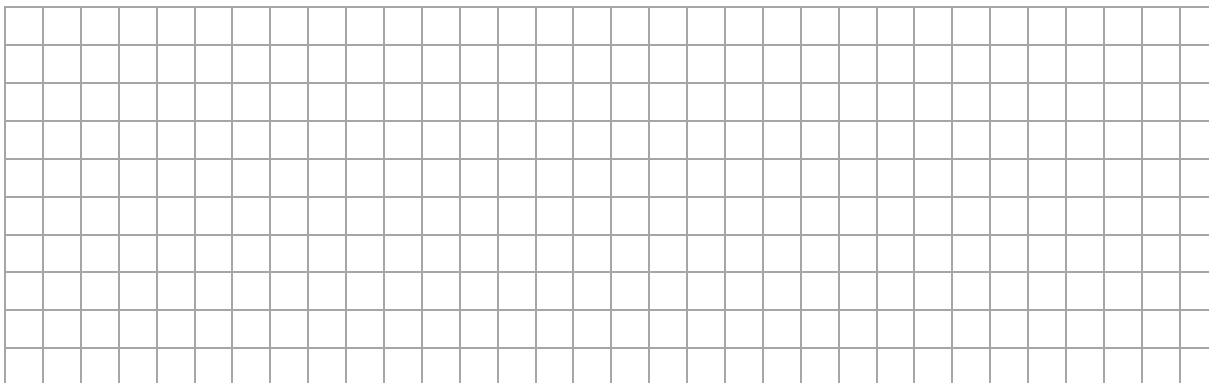
- Justification:

- (d) On the co-ordinated diagram, draw the two diagonals of $ABCD$.
Write down the co-ordinates of M , the point of intersection of these diagonals.

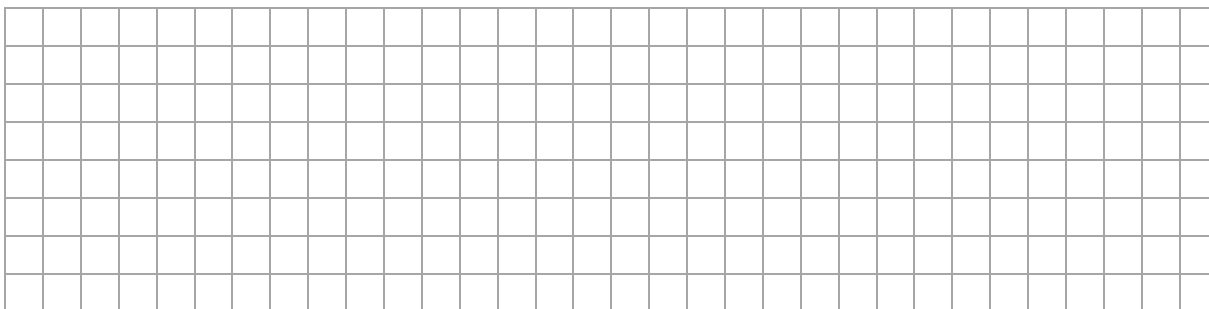
$$M = \left(\quad , \quad \right)$$

- (e) Work out $|AC|$, the **length** of $[AC]$.
Give your answer correct to one decimal place. Show your working out.

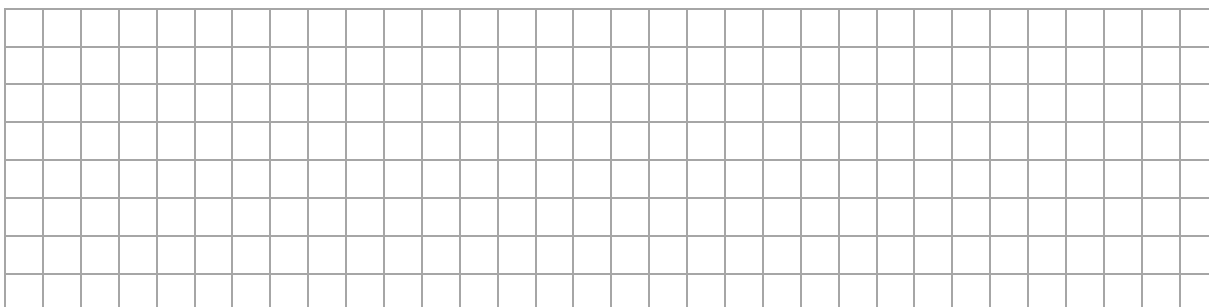
$$|AC| = \boxed{}$$



- (f) Find the **slope** of the line BC .



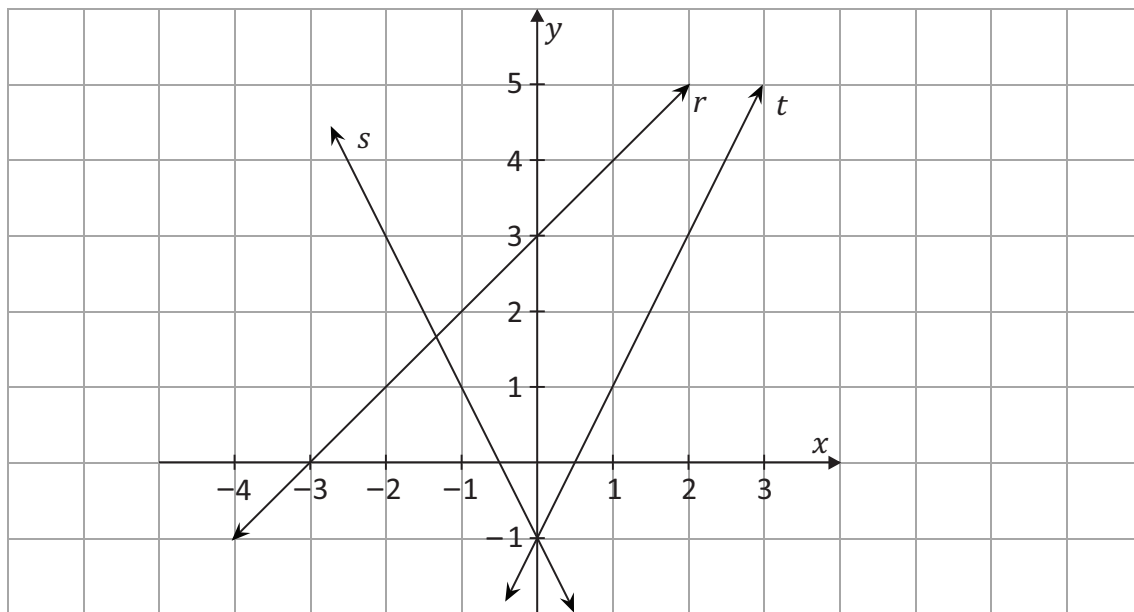
- (g) Use your answer to part (f) to work out the equation of the line BC .
Write your answer in the form $y = mx + c$.



Question 8 (Suggested maximum time: 5 minutes)

Question 8 (Suggested maximum time: 5 minutes)

The co-ordinate diagram below shows the lines r , s and t .



- (a) The table below shows the slopes and equations of the lines r , s and t . Fill in the three missing entries.

Line	Slope	Equation
r		$y = x + 3$
s		$y = -2x - 1$
t	2	

[illegible]

- (b)** Is the point $(5, -12)$ on the line s ($y = -2x - 1$)? **Justify** your answer.

The point $(5, -12)$

is on s

is not on s

Tick (✓) **one** box only.

7

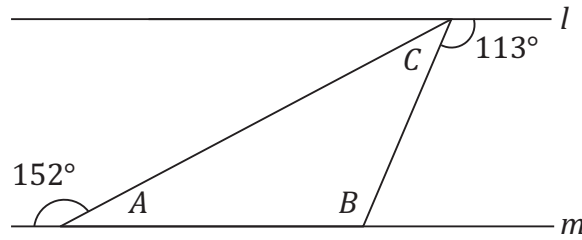
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- Justification:

[illegible]

(Suggested maximum time: 10 minutes)

In the diagram below, the line l is parallel to the line m .
The size of some of the angles are marked



- (a)** Theorems on your course can be used to find the measure of the missing angles in the diagram above. Write down, in your own words, any theorem that you could use to find one of the missing angles.

[illegible]

- (b)** Work out the size of the angle A , the angle B and the angle C .

$$|\angle A| = \boxed{} \qquad |\angle B| = \boxed{} \qquad |\angle C| = \boxed{}$$

- (c) What type of triangle is shown in the diagram above? Tick (✓) **one** box only.
Give a reason for your answer.

scalene

1

7

7

Reason:

[illegible]

Question 10

(Suggested maximum time: 10 minutes)

- (a)** June is carrying out a survey into the amount of chocolate people eat in a day. The following is a question from her survey:



Too much chocolate is bad for your health.
How many pieces of chocolate did you eat yesterday?
Tick (✓) **one** box only.

1-2 ☐3-4 ☐5 - 6

- (i) Explain why this question is **biased**.

[illegible]

- (ii)** Explain one other reason why this question is **not** suitable.

[illegible]

- (iii) Rewrite the question in a suitable form.

[illegible]

- (iv)** Put a tick in the correct box to show what type of data this is.
Justify your answer. Tick (✓) **one** box only.

Categorical

7

Numerical

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- Justification:

[illegible]

- (b)** Jacob asked everyone in his class to donate all the coins in their pocket to a charity. The stem and leaf diagram below shows the value of coins received from each student.

3	6	7						
4	4	4	6					
5	6	6	6	7	9			
6	1	4	8					
7	3	6						
8	4	5						

Key:	3	6	=	36
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- (i) Write down the **mode** of the data.

Answer:

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- (ii) Write down the **median** of the data.

Answer:

[illegible]

- (iii) Work out the **range** of the data.

[illegible]

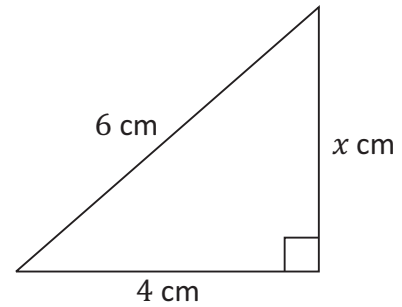
- (iv) The **total value** of all the coins is €10.02.
Work out the **mean** value of coins donated by each student.
Give your answer correct to the nearest cent.

[illegible]

Question 11

(Suggested maximum time: 15 minutes)

The diagram shows a right-angled triangle.
The length of the hypotenuse is 6 cm.
The lengths of the other two sides are 4 cm
and x cm as shown.
The diagram is **not** to scale.



- (a) **Construct** this triangle in the space below.
Show all of your construction lines clearly.

Construction:

- (b)** Barry says: “the **hypotenuse** is always the longest side in a right-angled triangle.”
Louise says: “the **hypotenuse** is always opposite to the right-angle in a right-angled triangle.”
Is only one or are both statements correct? Tick (✓) **one** box only.
Give a reason for your answer.

Only Barry is correct

1

Only Louise is correct

7

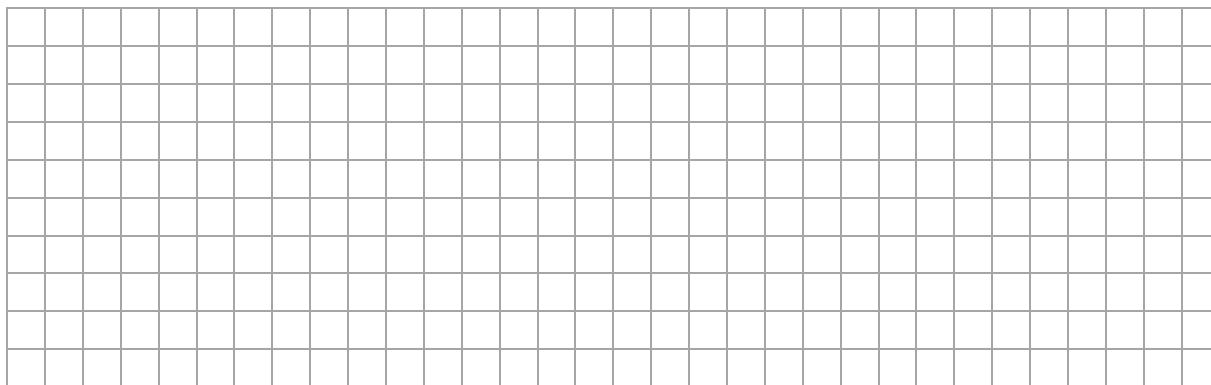
Both are correct

7

Reason:

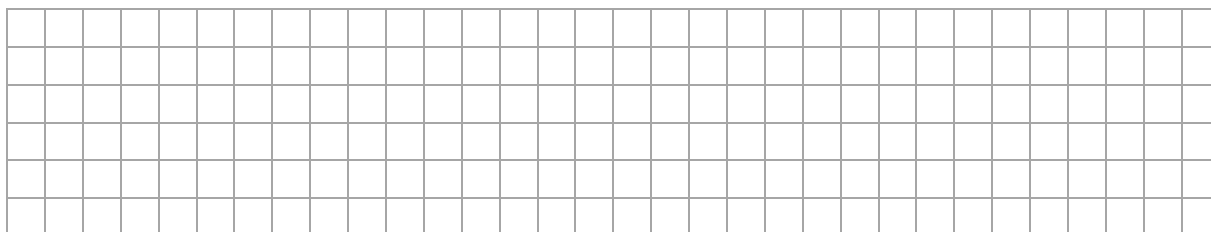
[illegible]

- (c) Use the theorem of **Pythagoras** to find the value of x , the third side of the triangle.
Give your answer in cm, correct to one decimal place.



- (d) Work out the **area** of the triangle.
Give your answer in cm^2 , correct to one decimal place.

Area:

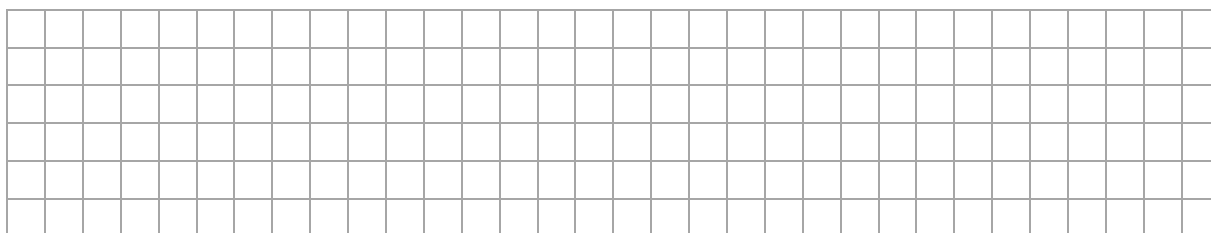


- (e) A is an angle in the triangle with

$$\cos A = \frac{4}{6}.$$

Use this fact to work out the **size** of the angle A , correct to the nearest degree.

Answer:

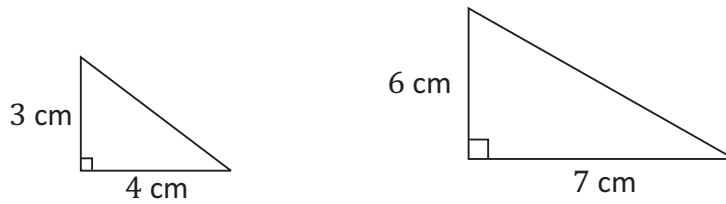


- (f) Show the location of the angle A in the triangle you constructed in part (a).

Question 12

(Suggested maximum time: 5 minutes)

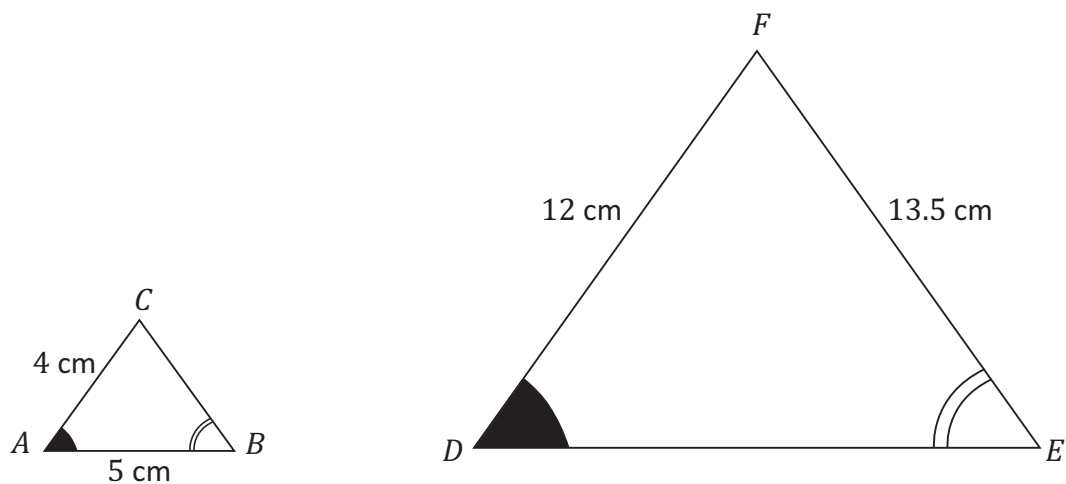
- (a)** The diagram below shows two right angled triangles.
Penny says: “the two triangles are **similar** because 3 cm was added to both the height and base of the smaller triangle to make the larger triangle.”



Explain why Penny is **not** correct.

[illegible]

- (b)** Two similar triangles are shown in the diagrams below.



- (i) Work out the length of $[DE]$.

[illegible]

- (ii) Work out the length of $[BC]$.

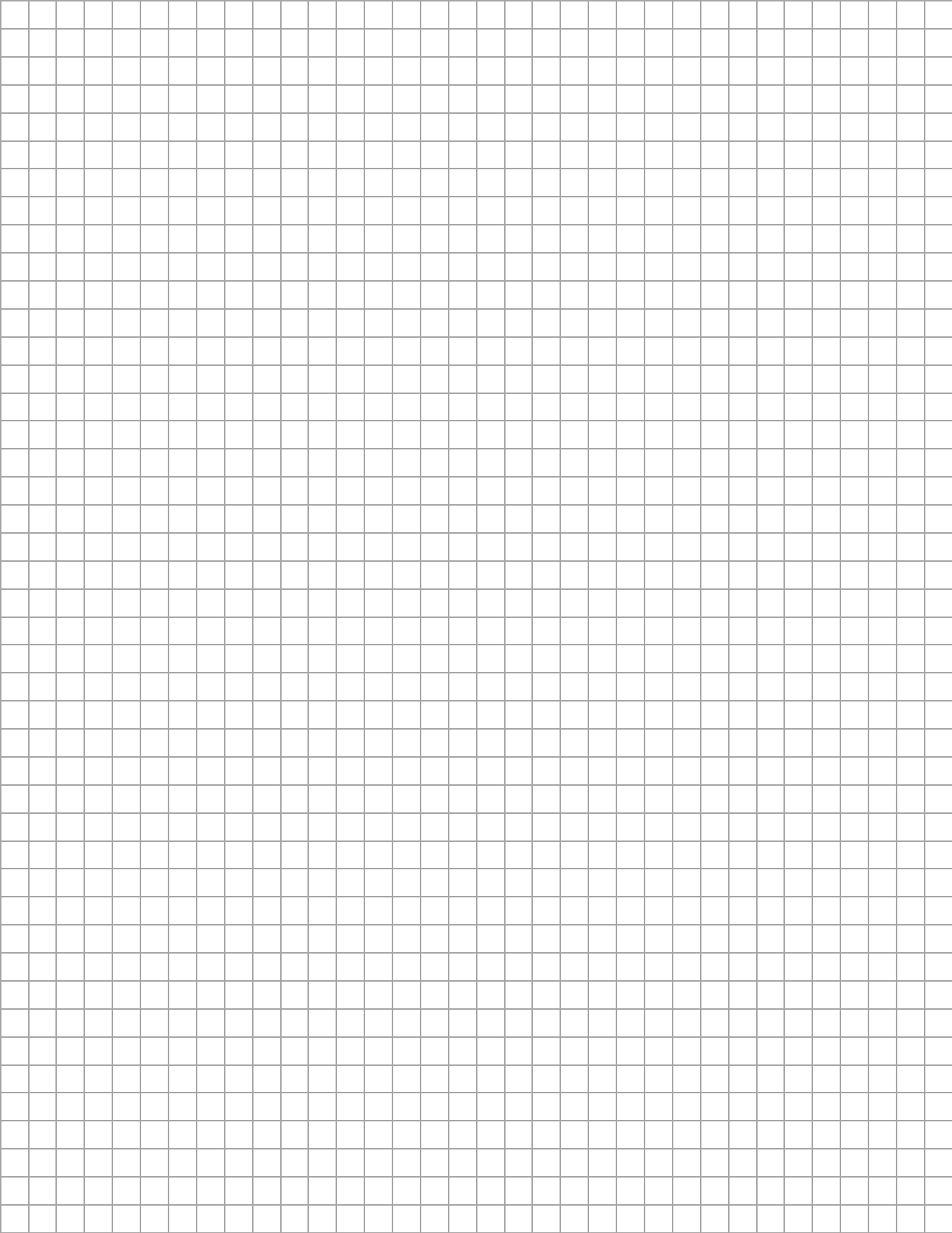
[illegible]

Page for extra work.

Label any extra work clearly with the question number and part.

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

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Pre-Junior Certificate Examination, 2020

Mathematics – Paper 2

Ordinary Level

Time: 2 hours

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